

Technical summary: the unrestricted Lalonde coloring

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Result. Let $G_n = G_{19} \vee K_{n-3}$ for $n \geq 3$, with G_{19} given by graph6 string `RxLAKA@AgYAWDGO?0?@??A?W@00C@_`. In the unrestricted finite-dimensional projector model,

$$\boxed{\chi_q(G_n) = n + 1 \quad (n \geq 3)}.$$

In particular, for the one-apex graph $H = G_4$, $\chi_q(H) = 5$. Zero projectors, nonuniform original ranks, arbitrary finite dimension, and noncommuting projectors are all allowed.

Uniformization and clique saturation. Suppose G_n has a quantum n -coloring in dimension d . Direct-sum over all color permutations:

$$\tilde{P}_{v,c} = \bigoplus_{\pi \in S_n} P_{v,\pi(c)}.$$

The new dimension is $D = n!d = nr$, and every projector has common positive rank $r = (n-1)!d$. Thus uniformity is a theorem-level reduction, not an ansatz. For an n -clique, each fixed-color column consists of pairwise orthogonal projections. Its trace is at most D ; summing over all colors and using the row partitions makes all n inequalities equalities. Hence every clique column sums to I_D .

Fix a color c . Let R_c be the sum of its projections on the joined clique and $Q_c = I - R_c$. Each of 123, 189, 345, and 267, joined with K_{n-3} , is an n -clique. Therefore

$$\dim Q_c = 3r, \quad \sum_{v \in T_j} P_{v,c} = Q_c.$$

On Q_c , write $e_v = P_{v,c}|_{Q_c}$. These are rank- r projections giving a projective orthogonal representation of G_{19} .

Exact core SOS. Set $X = e_{10} + e_{11} + e_{12} + e_{13}$,

$$\begin{aligned} H_1 &= e_{10} + e_{11} - e_{12} - e_{13}, \\ H_2 &= e_{10} - e_{11} + e_{12} - e_{13}, & (d_1, d_2, d_3) &= (e_5 - e_4, e_7 - e_6, e_9 - e_8). \\ H_3 &= e_{10} - e_{11} - e_{12} + e_{13}, \end{aligned}$$

Projection, edge, and triangle relations give

$$\sum_i d_i^2 = 2I, \quad X^2 + \sum_i H_i^2 = 4X, \quad \sum_i \{d_i, H_i\} = -4X.$$

The final identity is graph-specific: for each $j = 10, \dots, 13$, the positive Walsh signs select exactly the three neighbors of j among $4, \dots, 9$. Expanding the following factors now gives the exact rational identity

$$\frac{4}{3}I - X = \sum_{j=0}^3 f_j^* f_j, \quad f_0 = \frac{1}{2} \left(X - \frac{4}{3}I \right), \quad f_i = \frac{2}{3}d_i + \frac{1}{2}H_i.$$

Both sides have trace zero, since $\dim Q_c = 3r$ and each e_j has rank r . Faithfulness of finite matrix trace forces every $f_j = 0$.

Put $A = e_4 - e_5$, $B = e_6 - e_7$, and $C = e_8 - e_9$. Walsh inversion and $e_j^2 = e_j$ yield

$$\{B, C\} = A, \quad \{A, C\} = B, \quad \{A, B\} = C.$$

Relative to $Q_c = \text{ran } e_1 \oplus \text{ran } e_2 \oplus \text{ran } e_3$, support restrictions put A, B, C on coordinate pairs 12, 13, 23. The anticommutator equations kill their diagonal blocks; the remaining blocks are unitaries with one cocycle relation. A block-diagonal gauge makes all three blocks I_r . Thus vertices $1, \dots, 13$ are exactly the scalar sign-vector core tensored with a multiplicity space $K_c \cong \mathbb{C}^r$.

Exhaustive tail. The remaining ranges are encoded by an arbitrary r -plane $M_c \subset K_c \oplus K_c$ invariant under $J_c = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$:

$$\begin{array}{c|ccc} v & 14 & 15 & 16 \\ \hline \text{ran } e_v & (a, 0, b) & (0, a, b) & (a, -b, 0) \end{array} \quad (a, b) \in M_c,$$

$$\begin{array}{c|ccc} v & 17 & 18 & 19 \\ \hline \text{ran } e_v & (x, x, z) & (x, -z, z) & (x, -z, x) \end{array} \quad (x, z) \in M_c^\perp.$$

The six core–tail edges supply the displayed ambient frames. Pulling back the six tail–tail edges gives five identity maps and one $-J_c$, proving both exhaustiveness and $J_c M_c = M_c$. This formulation includes non-graph planes; $M_c = \text{ran}(I, S_c)$ with $S_c^2 = -I$ is only a transverse chart.

Cross-color sector packing. Choose a core unitary $W_c : K_c^3 \rightarrow Q_c$. For $c \neq d$, same-vertex orthogonality at vertices $1, \dots, 9$ forces

$$W_d^* W_c = \begin{pmatrix} 0 & X & Y \\ -X & 0 & Z \\ -Y & -Z & 0 \end{pmatrix}.$$

The lower blocks are negatives, not adjoints, because the matrix maps the c coordinate space to the d coordinate space. With $\Omega_L = \begin{pmatrix} 0 & L \\ -L & 0 \end{pmatrix}$, the six tail compressions are

$$\Omega_Y, \Omega_Z, -\Omega_X, \Omega_{Y+Z}, \Omega_{Y-X}, \Omega_{Z-X}.$$

Their invertible coefficient transform implies, for $L = X, Y, Z$,

$$\Omega_L M_c \subseteq M_d^\perp, \quad \Omega_L M_c^\perp \subseteq M_d.$$

Since M_c reduces J_c , write

$$M_c \cap \ker(J_c - \sigma i I) = \{(x, \sigma i x) : x \in E_{c, \sigma}\}, \quad e_{c, +} + e_{c, -} = r.$$

Each Ω_L intertwines J_c and J_d , so the plane flip says $LE_{c, \sigma} \subseteq E_{d, \sigma}^\perp$. For fixed σ , the n physical spaces

$$W_c(E_{c, \sigma} \oplus E_{c, \sigma} \oplus E_{c, \sigma})$$

are pairwise orthogonal in dimension nr . Hence

$$3 \sum_c e_{c, +} \leq nr, \quad 3 \sum_c e_{c, -} \leq nr.$$

Adding gives $3nr \leq 2nr$, impossible for $r > 0$. Thus no quantum n -coloring exists. The explicit classical four-coloring of G_{19} and fresh colors on the joined clique give the matching $(n+1)$ upper bound.

Verification. Two standard-library exact verifiers accompany the proof. They decode the graph6 checksum, replay independent rational noncommutative SOS expansions, check the Walsh and skew-block systems, multiply all six tail compressions, invert their coefficient transform, and check the symbolic all- n dimension gap. No floating-point computation enters the theorem.

Scope and relation to prior work. The known rank-one theorem motivates the sign frame, but it is not used as a reduction: the SOS proves the module-valued core directly, and the invariant plane analysis handles genuinely higher-rank tails. The contradiction does not require irreducibility and therefore excludes every finite-dimensional representation of the n -coloring relations, reducible or irreducible. No minimal forbidden local dimension exists. At the valid $(n+1)$ -color level, the explicit classical coloring is already a dimension-one representation. The theorem makes no assertion about infinite-dimensional or commuting-operator colorings.